Stochastic Processes

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Contents

1 Introduction 1
1.1 Probability 1
1.2 Conditional Expectation 3
1.3 Stochastic Process 5
2 Poisson Process 6
2.1 Homogeneous Poisson Process 6
2.2 Arrival Time 8
2.3 Other Poisson Processes 10
3 Markov Chains 12
3.1 Discrete-Time Markov Chains 12
3.2 Transition Matrix 14
3.3 Classification of States 15

1 Introduction

1.1 Probability

Definition 1.1. *(Probability Space)*
A probability space is a triple $(\Omega, \mathcal{F}, P)$. The sample space Ω is the set of outcomes. The event field $\mathcal{F}$ is a σ-field of subsets of Ω:

- $\Omega \in \mathcal{F}$;
- $A \in \mathcal{F}$ implies $A^c \in \mathcal{F}$;
- $A_k \in \mathcal{F}$ implies $\bigcup_{k=1}^{\infty} A_k \in \mathcal{F}$.

Definition 1.2. *(Probability)*
Given a probability space $(\Omega, \mathcal{F}, P)$, a set function $P : \mathcal{F} \to \mathbb{R}$ is a probability if

- $P(A) \geq 0$ for every $A \in \mathcal{F}$;
- $P(\Omega) = 1$;
- whenever $A_i \in \mathcal{F}$ are pairwise disjoint, $P\left(\bigcup_i A_i\right) = \sum_i P(A_i)$.

Definition 1.3. *(Random Variable)*
Given a probability space $(\Omega, \mathcal{F}, P)$, a map $X : \Omega \to \mathbb{R}$ is a random variable if $\{\omega : X(\omega) \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$.

Definition 1.4. *(Distribution Function)*
Given a random variable X, the distribution function is $F_X(x) = P(X \leq x)$.

Definition 1.5. *(Expectation)*
If the integral exists,

$$E(X) = \int_{-\infty}^{+\infty} x \, \mathrm{d}F_X(x).$$

This Riemann-Stieltjes integral is $\sum_i x_i(F_X(x_i) - F_X(x_i -))$ in the discrete case and $\int x f_X(x) \, \mathrm{d}x$ in the continuous case. For a measurable g,

$$E(g(X)) = \int g(x) \, \mathrm{d}F_X(x).$$

Proposition 1.1.
Expectation is linear whether or not the summands are independent.

Definition 1.6. *(Variance, Covariance, and Correlation)*
The variance, covariance, and correlation are

$$\mathrm{Var}(X) = E(X - EX)^2 = E(X^2) - (EX)^2,$$

$$\mathrm{Cov}(X, Y) = E((X - EX)(Y - EY)) = E(XY) - E(X)E(Y),$$

$$\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\ \mathrm{Var}(Y)}}.$$

Proposition 1.2.
$|E(XY)| \leq \sqrt{E(X^2)E(Y^2)}$, and therefore $|\mathrm{Cov}(X, Y)| \leq \sqrt{\mathrm{Var}(X)\ \mathrm{Var}(Y)}$.

1.2 Conditional Expectation

Definition 1.7. *(Conditional Expectation)*
Let $E|X| < \infty$. A random variable $E(X|Y)$ is a conditional expectation of X given Y if $E(X|Y) = g(Y)$ for some measurable g with $E|g(Y)| < \infty$, and

$$E(g(Y)I_D(Y)) = E(XI_D(Y))$$

for every Borel set D.

Proposition 1.3.
- $E(E(X|Y)) = E(X)$.
- If $h(Y)$ is bounded, then $E(g(X)h(Y) \mid Y) = h(Y)E(g(X)|Y)$. In particular $E(E(X|Y)h(Y)) = E(Xh(Y))$.
- If X and Y are independent, then $E(X|Y) = E(X)$.

Remark.
The proposition says that $X - E(X|Y)$ is orthogonal to every bounded function of Y: $\operatorname{Cov}(X - E(X|Y), h(Y)) = 0$ whenever the covariance exists.

Theorem 1.1. *(Least Squares)*
If $E(X^2) < \infty$ and $g(Y)$ is square-integrable, then

$$E(X - g(Y))^2 \geq E(X - E(X|Y))^2.$$

Proof.
Write $X - g(Y) = (X - E(X|Y)) + (E(X|Y) - g(Y))$. In the expanded square, the cross term vanishes because $E(X|Y) - g(Y)$ is a function of Y and is orthogonal to $X - E(X|Y)$. What remains is

$$E(X - E(X|Y))^2 + E(E(X|Y) - g(Y))^2,$$

and the second summand is nonnegative. ■

Theorem 1.2. *(Independence)*
If X and Y are independent and $E|h(X, Y)| < \infty$, then $E(h(X, Y)|Y) = g(Y)$, where $g(y) = E(h(X, y))$.

Proof.
Independence gives $\mathrm{d}F_{X,Y} = \mathrm{d}F_X \, \mathrm{d}F_Y$. For every Borel set D,

$$E(h(X,Y)I_D(Y)) = \int\int h(x,y)I_D(y)\,\mathrm{d}F_X(x)\,\mathrm{d}F_Y(y) = E(g(Y)I_D(Y)).$$

■

Theorem 1.3. *(Tower Property)*
If $1 \le m \le n$ and $E|X| < \infty$, then

$$E(X \mid Y_1, ..., Y_m) = E(E(X \mid Y_1, ..., Y_n) \mid Y_1, ..., Y_m).$$

Proof.
It is enough to show that $X - E(X|Y_1, ..., Y_n)$ has conditional expectation zero given $(Y_1, ..., Y_m)$. For every bounded measurable $h(y_1, ..., y_m)$, the product $h(Y_1, ..., Y_m)$ is a bounded function of $(Y_1, ..., Y_n)$, so

$$E(Xh(Y_1, ..., Y_m)) = E(E(X|Y_1, ..., Y_n)h(Y_1, ..., Y_m)).$$

The two sides therefore agree, and the difference $X - E(X|Y_1, ..., Y_n)$ is orthogonal to every bounded function of $(Y_1, ..., Y_m)$. ■

Definition 1.8. *(Conditional Variance)*

$$\mathrm{Var}(Y|X) = E((Y - E(Y|X))^2 \mid X).$$

Theorem 1.4. *(Law of Total Variance)*
If $E(Y^2) < \infty$, then

$$\mathrm{Var}(Y) = E(\mathrm{Var}(Y|X)) + \mathrm{Var}(E(Y|X)).$$

In particular $\mathrm{Var}(Y) \ge E(\mathrm{Var}(Y|X)), E(\mathrm{Var}(Y|X_1)) \ge E(\mathrm{Var}(Y|X_1, X_2))$.

Proof.

$$\mathrm{Var}(Y) = E(Y^2) - (EY)^2 = E(E(Y^2|X)) - (E(E(Y|X)))^2.$$

Substitute $E(Y^2|X) = \mathrm{Var}(Y|X) + (E(Y|X))^2$. The same decomposition given X_1, applied to the conditioning σ-field of (X_1, X_2), yields

$$\mathrm{Var}(Y|X_1) = E(\mathrm{Var}(Y|X_1, X_2)|X_1) + \mathrm{Var}(E(Y|X_1, X_2)|X_1).$$

The second summand is nonnegative. Taking expectations gives the comparison. ■

Theorem 1.5. *(Dominated Convergence)*
Let $EY < \infty$ with $Y \geq 0$, and suppose $|X_n| \leq Y$ almost surely. If $X_n \overset{a.s.}{\to} X$, then X is integrable and

$$E(X_n) \overset{a.s.}{\to} E(X).$$

If instead $X_n \overset{P}{\to} X$, the same convergence holds in probability.

1.3 Stochastic Process

Definition 1.9. *(Stochastic Process)*
Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $\mathbb{T}$ be a time set. A stochastic process is a family $X = \{X_t : t \in \mathbb{T}\}$ such that X_t is a random variable for every $t \in \mathbb{T}$.

Definition 1.10. *(Finite-Dimensional Distributions)*
For $t_1 < ... < t_n$ in $\mathbb{T}$, the distribution function

$$F_{t_1,...,t_n}(x_1, ..., x_n) = P\Big(X_{t_1} \leq x_1, ..., X_{t_n} \leq x_n\Big)$$

is a finite-dimensional distribution of X. Every process has one, and it is unique.

Proposition 1.4. *(Symmetry and Consistency)*
The finite-dimensional distribution of a process satisfy

- symmetry: for every permutation $(j_1, ..., j_n)$ of $(1, ..., n)$,

$$F_{t_{j_1},...,t_{j_n}}\left(x_{j_1}, ..., x_{j_n}\right) = F_{t_1,...,t_n}(x_1, ..., x_n);$$

- consistency: if $m < n$, then

$$F_{t_1,...,t_n}(x_1, ..., x_m, +\infty, ..., +\infty) = F_{t_1,...,t_m}(x_1, ..., x_m).$$

Definition 1.11. *(Mean, Covariance, and Correlation Functions)*
For a real process $\{X_t : t \in \mathbb{T}\}$,

$$m(t) = E(X_t), \quad \mathrm{Var}(X_t) = E(X_t - m(t))^2,$$

$$R(s,t) = E(X_s X_t), \quad C_X(s,t) = \mathrm{Cov}(X_s, X_t) = R(s,t) - m(s)m(t).$$

The correlation function of the process is R, and the correlation coefficient is

$$\rho(s,t) = \frac{C_X(s,t)}{\sqrt{\operatorname{Var}(X_s)\,\operatorname{Var}(X_t)}}.$$

2 Poisson Process

2.1 Homogeneous Poisson Process

Definition 2.1. *(Counting Process)*
A counting process $\{N_t : t \geq 0\}$ satisfies $N_0 = 0$, takes values in $\{0, 1, 2, ...\}$, and has nondecreasing paths. Then N_t counts events in $[0, t]$, and $N_t - N_s$ counts events in $(s, t]$.

Definition 2.2. *(Independent and Stationary Increments)*
A process X has independent increments if, for every $t_0 < t_1 < ... < t_n$, the increments $X_{t_1} - X_{t_0}, ..., X_{t_n} - X_{t_{n-1}}$ are independent. It has stationary increments if, for $0 \leq s < t$, the increment $X_t - X_s$ has the same distribution as $X_{t-s} - X_0$.

Definition 2.3. *(Homogeneous Poisson Process)*
A counting process $\{N_t : t \geq 0\}$ is a homogeneous Poisson process of intensity $\lambda > 0$ if

- it has independent increments and stationary increments;
- for every $t > 0$ and every sufficiently small $h > 0$,

$$P(N_{t+h} - N_t = 1) = \lambda h + o(h),$$

$$P(N_{t+h} - N_t = 0) = 1 - \lambda h + o(h).$$

Theorem 2.1.
If $\{N_t : t \geq 0\}$ is a homogeneous Poisson process of intensity λ, then for every $s \geq 0$ and $t > 0$,

$$N_{s+t} - N_s \sim \text{Poisson}(\lambda t).$$

In particular, $P(N_t = k) = e^{-\lambda t}\frac{(\lambda t)^k}{k!}$ for $k = 0, 1, 2, ...$

Proof.
Write $p_k(t) = P(N_t = k)$. Independent and stationary increments give, for $h > 0$,

$$p_0(t+h) = p_0(t)(1-\lambda h + o(h)).$$

Dividing by h and letting $h \to 0$ yields $p_0'(t) = -\lambda p_0(t)$. Since $N_0 = 0$, one has $p_0(0) = 1$, so $p_0(t) = e^{-\lambda t}$.
For $k \geq 1$ the same splitting, and the estimate $P(\text{increment} \geq 2) = o(h)$, gives

$$p_k(t+h) = p_k(t)(1-\lambda h + o(h)) + p_{k-1}(t)(\lambda h + o(h)) + o(h),$$

hence $p_k'(t) = -\lambda p_k(t) + \lambda p_{k-1}(t)$.
The probability generating function $\Phi(t,z) = \sum_{k=0}^{\infty} p_k(t) z^k$ satisfies $\Phi(0,z) = 1$ and

$$\partial_t \Phi(t,z) = -\lambda(1-z)\Phi(t,z).$$

Therefore $\Phi(t,z) = \exp(-\lambda t(1-z)) = e^{-\lambda t}\sum_{k=0}^{\infty} (\lambda t)^k \frac{z^k}{k!}$, and the coefficient of z^k is the Poisson probability. ■

Definition 2.4. *(Second Definition)*
A counting process $\{N_t : t \geq 0\}$ is a homogeneous Poisson process of intensity λ if $N_0 = 0$, the process has independent increments, and $N_{s+t} - N_s \sim \text{Poisson}(\lambda t)$ for all $s, t \geq 0$.

Proposition 2.1.
If $\{N_t\}$ is a homogeneous Poisson process of intensity λ, then $E(N_t) = \text{Var}(N_t) = \lambda t$ and

$$\text{Cov}(N_s, N_t) = \lambda \min(s,t).$$

Proof.
The mean and variance are those of $\text{Poisson}(\lambda t)$. If $s \leq t$, independent increments give

$$\text{Cov}(N_s, N_t) = \text{Cov}(N_s, N_s + (N_t - N_s)) = \text{Var}(N_s) = \lambda s.$$

■

Theorem 2.2.
If $N^{(1)}, \ldots, N^{(m)}$ are independent homogeneous Poisson processes of intensities $\lambda_1, \ldots, \lambda_m$, then $N_t = N_t^{(1)} + \ldots + N_t^{(m)}$ is a homogeneous Poisson process of intensity $\lambda_1 + \ldots + \lambda_m$.

Theorem 2.3.
Let $\{N_t\}$ be a homogeneous Poisson process of intensity λ. Mark each event independently as type 1 with probability p and type 2 with probability $1-p$, and let $N_t^{(1)}$ and $N_t^{(2)}$ be the counts of the two types in $[0,t]$. Then for each fixed t,

$$N_t^{(1)} \sim \text{Poisson}(\lambda p t), \quad N_t^{(2)} \sim \text{Poisson}(\lambda(1-p)t),$$

and $N_t^{(1)}$ is independent of $N_t^{(2)}$.

Proof.
Condition on the total count and use the binomial thinning:

$$P\Big(N_t^{(1)} = i, N_t^{(2)} = j\Big) = e^{-\lambda t}\frac{(\lambda t)^{i+j}}{(i+j)!} \cdot \binom{i+j}{i} p^i (1-p)^j,$$

which factors as the product of the two Poisson probabilities. ■

2.2 Arrival Time

Definition 2.5. *(Arrival and Interarrival Time)*
Let $\tau_0 = 0$ and let τ_n be the time of the n-th event. The interarrival time is $X_n = \tau_n - \tau_{n-1}$ for $n \geq 1$.

Theorem 2.4.
If $\{N_t\}$ is a homogeneous Poisson process of intensity λ, then $\tau_n \sim \Gamma(n, \lambda)$:

$$f_{\tau_n}(t) = \frac{\lambda^n}{(n-1)!} t^{n-1} e^{-\lambda t}, \quad t \geq 0.$$

Proof.

$$P(\tau_n \leq t) = P(N_t \geq n) = \sum_{k=n}^{\infty} e^{-\lambda t}\frac{(\lambda t)^k}{k!}.$$

Differentiating term by term,

$$\frac{d}{dt}\left(e^{-\lambda t}\frac{(\lambda t)^k}{k!}\right) = -\lambda e^{-\lambda t}\frac{(\lambda t)^k}{k!} + \lambda e^{-\lambda t}\frac{(\lambda t)^{k-1}}{(k-1)!}.$$

Summing from $k = n$ to ∞ and reindexing the second sum yields the density. ■

Theorem 2.5.
On $0 < t_1 < \ldots < t_n < \infty$, the joint density of $(\tau_1, \ldots, \tau_n)$ is $\lambda^n e^{-\lambda t_n}$.

Proof.
If $h > 0$ is small enough that the windows $\left(t_i - \frac{h}{2}, t_i + \frac{h}{2}\right)$ are disjoint, the probability of one event in each window and none elsewhere on $\left[0, t_n + \frac{h}{2}\right]$ equals $(\lambda h)^n e^{-\lambda\left(t_n + \frac{h}{2}\right)}$. Dividing by h^n and letting $h \to 0^+$ yields the density $\lambda^n e^{-\lambda t_n}$. ■

Theorem 2.6.
A counting process is a homogeneous Poisson process of intensity λ if and only if its interarrival times $X_i \sim \mathrm{Exp}(\lambda)$ i.i.d..

Proof.
Suppose N is Poisson of intensity λ. The map $(x_1, \ldots, x_n) \mapsto (x_1, x_1 + x_2, \ldots, x_1 + \ldots + x_n)$ has Jacobian determinant 1. Substituting $t_n = x_1 + \ldots + x_n$ into the joint density of the arrival times gives, for $x_i > 0$,

$$f_{X_1,\ldots,X_n}(x_1, \ldots, x_n) = \lambda^n e^{-\lambda(x_1 + \ldots + x_n)} = \prod_{i=1}^{n} \lambda e^{-\lambda x_i}.$$

Thus the interarrival times are i.i.d. exponential of rate λ.
Conversely, start from i.i.d. exponential interarrival times, set $\tau_n = X_1 + \ldots + X_n$ and $N_t = \sup\{n : \tau_n \le t\}$. Then $\tau_n \sim \Gamma(n, \lambda)$, so

$$P(N_t = n) = P(\tau_n \le t) - P(\tau_{n+1} \le t) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}.$$

The memoryless property of the exponential law makes the time until the next event, after any fixed time, exponential of rate λ and independent of the history. ■

Theorem 2.7.
If $\{N_t\}$ is a homogeneous Poisson process of intensity λ and $n \ge 1$, then conditionally on $N_t = n$ the arrival times $(\tau_1, \ldots, \tau_n)$ have density

$$f(t_1, \ldots, t_n \mid N_t = n) = \frac{n!}{t^n}, \quad 0 < t_1 < \ldots < t_n < t.$$

Namely, they are distributed as the order statistics of n i.i.d. uniform random variables on $[0, t]$.

Proof.
For the joint density, take disjoint windows of width h about $t_1 < \ldots < t_n$. The

probability of one event in each window and none elsewhere in $[0,t]$ is $e^{-\lambda t}(\lambda h)^n$, and $P(N_t = n) = e^{-\lambda t}\frac{(\lambda t)^n}{n!}$. The ratio is $n!\left(\frac{h}{t}\right)^n$. Dividing by h^n produces $\frac{n!}{t^n}$. ■

Example.
Passengers arrive at a stop as a homogeneous Poisson process of intensity λ per minute. A bus clears the queue every 10 minutes. Let τ_k be the arrival time of the k-th passenger during a ten-minute cycle, and let $X = \sum_{k=1}^{N_{10}} (10 - \tau_k)$ be the total waiting time in that cycle. Then

$$E(X) = E(E(X \mid N_{10})) = E(5N_{10}) = 50\lambda.$$

2.3 Other Poisson Processes

Definition 2.6. *(Nonhomogeneous Poisson Process)*
Let $\lambda(t) > 0$ be locally integrable. A counting process $\{N_t : t \geq 0\}$ is a nonhomogeneous Poisson process of intensity $\lambda(t)$ if it has independent increments and for every $t \geq 0$ and every sufficiently small $h > 0$,

$$P(N_{t+h} - N_t = 1) = \lambda(t)h + o(h),$$

$$P(N_{t+h} - N_t = 0) = 1 - \lambda(t)h + o(h).$$

Theorem 2.8.
A counting process is a nonhomogeneous Poisson process of intensity $\lambda(t)$ if and only if it has independent increments and

$$N_t - N_s \sim \text{Poisson}(m(t) - m(s)), \quad 0 \leq s < t,$$

where $m(t) = \int_0^t \lambda(u)\,\mathrm{d}u$.

Definition 2.7. *(Compound Poisson Process)*
Let $\{\xi_i : i = 1, 2, \ldots\}$ be i.i.d. random variables, independent of a homogeneous Poisson process $\{N_t : t \geq 0\}$ of intensity λ. The process

$$X_t = \sum_{i=1}^{N_t} \xi_i,$$

with the empty sum equal to zero, is a compound Poisson process.

Proposition 2.2.
If $\{X_t\}$ is compound Poisson as above, with $\mu = E(\xi_1)$ and $E(\xi_1^2) < \infty$, then

$$E(X_t) = \lambda t \mu, \quad \operatorname{Var}(X_t) = \lambda t E(\xi_1^2),$$

$$R(s,t) = E(X_s X_t) = \lambda \min(s,t) E(\xi_1^2) + \lambda^2 s t \mu^2,$$

$$\operatorname{Cov}(X_s, X_t) = \lambda \min(s,t) E(\xi_1^2).$$

Proof.
Conditioning on the count gives $E(X_t \mid N_t) = N_t \mu$, hence $E(X_t) = E(N_t)\mu = \lambda t \mu$. For the correlation, take $s \le t$. Given $N_s = m$ and $N_t = n$, necessarily $m \le n$, and

$$E(X_s X_t \mid N_s = m, N_t = n) = \sum_{i=1}^{m} \sum_{j=1}^{n} E(\xi_i \xi_j) = m E(\xi_1^2) + (mn - m)\mu^2$$

$$= m \operatorname{Var}(\xi_1) + mn\mu^2.$$

Therefore

$$E(X_s X_t) = E(N_s) \operatorname{Var}(\xi_1) + E(N_s N_t)\mu^2.$$

Independent increments of N yield $E(N_s N_t) = \lambda s + \lambda^2 s t$, so

$$R(s,t) = \lambda s \operatorname{Var}(\xi_1) + (\lambda s + \lambda^2 s t)\mu^2 = \lambda s E(\xi_1^2) + \lambda^2 s t \mu^2.$$

The identities for general s, t follow by symmetry. Subtracting $E(X_s)E(X_t)$ leaves the covariance, and the variance is the case $s = t$. ■

Definition 2.8. *(Conditional Poisson Process)*
Let Λ be a nonnegative random variable with distribution function F_Λ. A counting process $\{N_t : t \ge 0\}$ is a conditional Poisson process with prior F_Λ if, given $\Lambda = \lambda$, the process N is a homogeneous Poisson process of intensity λ. Equivalently, for every $s < t$ and every $k = 0, 1, 2, \ldots$,

$$P(N_t - N_s = k \mid \Lambda = \lambda) = e^{-\lambda(t-s)} \frac{(\lambda(t-s))^k}{k!}.$$

Theorem 2.9. *(Gamma Prior)*
Suppose the prior of Λ is $\Gamma(m_0, \lambda_0)$, with density

$$f_\Lambda(\lambda) = \left(\frac{\lambda_0^{m_0}}{\Gamma(m_0)}\right)\lambda^{m_0-1}e^{-\lambda_0\lambda}, \quad \lambda > 0.$$

Conditionally on $N_1 = x_1$, $N_2 - N_1 = x_2$, ..., $N_n - N_{n-1} = x_n$, the posterior of Λ is $\Gamma(m_0 + x_1 + ... + x_n, n + \lambda_0)$.

Proof.
Independent increments given Λ yield

$$P(N_1 = x_1, ..., N_n - N_{n-1} = x_n \mid \Lambda = \lambda) = \prod_{i=1}^{n}\left(e^{-\lambda}\frac{\lambda^{x_i}}{x_{i!}}\right)$$

which is proportional to $\lambda^{x_1+...+x_n}e^{-n\lambda}$. The posterior density is therefore proportional to

$$\lambda^{x_1+...+x_n}e^{-n\lambda} \cdot \lambda^{m_0-1}e^{-\lambda_0\lambda} = \lambda^{m_0+x_1+...+x_n-1}e^{-(n+\lambda_0)\lambda},$$

which is the kernel of $\Gamma(m_0 + x_1 + ... + x_n, n + \lambda_0)$. ■

3 Markov Chains

3.1 Discrete-Time Markov Chains

Definition 3.1. *(Discrete-Time Markov Chain)*
The process is a discrete-time Markov chain if, whenever $P(X_0 = i_0, ..., X_n = i_n) > 0$,

$$P(X_{n+1} = i_{n+1} \mid X_0 = i_0, ..., X_n = i_n) = P(X_{n+1} = i_{n+1} \mid X_n = i_n).$$

Proposition 3.1.
The one-step Markov property is equivalent to the same identity at arbitrary times: for every $t_1 < ... < t_n < t_{n+1}$ with $P\left(X_{t_1} = i_1, ..., X_{t_n} = i_n\right) > 0$,

$$P\left(X_{t_{n+1}} = i_{n+1} \mid X_{t_1} = i_1, ..., X_{t_n} = i_n\right) = P\left(X_{t_{n+1}} = i_{n+1} \mid X_{t_n} = i_n\right).$$

Theorem 3.1.
The following are equivalent to the Markov property.

- For every bounded measurable f,

$$E(f(X_{n+1}) \mid X_0 = i_0, ..., X_n = i_n) = E(f(X_{n+1}) \mid X_n = i_n).$$

- For every $m \geq 1$,

$$P(X_{n+m} = i_{n+m} \mid X_0 = i_0, ..., X_n = i_n) = P(X_{n+m} = i_{n+m} \mid X_n = i_n).$$

- Write A for a past event $\{X_0 \in A_0, ..., X_{n-1} \in A_{n-1}\}$ and B for a future event $\{X_{n+1} \in A_{n+1}, X_{n+2} \in A_{n+2}, ...\}$. Then

$$P(B \mid A, X_n = i_n) = P(B \mid X_n = i_n).$$

Equivalently, past and future are conditionally independent given the present:

$$P(A \cap B \mid X_n = i_n) = P(A \mid X_n = i_n)P(B \mid X_n = i_n).$$

Definition 3.2. *(Transition Probability)*
The one-step transition probability at time n is

$$p_{ij}(n) = P(X_{n+1} = j \mid X_n = i).$$

The chain is homogeneous if $p_{ij}(n) = p_{ij}$ is independent of n. The matrix $\mathbb{P} = \left(p_{ij}\right)$ is then the transition matrix.

Theorem 3.2.
Suppose $X_{n+1} = f(X_n, \xi_{n+1})$ for $n \geq 0$, where $\{\xi_n\}$ is i.i.d. and independent of X_0. Then $\{X_n\}$ is a homogeneous Markov chain with

$$p_{ij} = P(f(i, \xi_1) = j).$$

Proof.
The vector $(X_0, ..., X_n)$ is a function of $(X_0, \xi_1, ..., \xi_n)$, hence independent of ξ_{n+1}. Therefore

$$P(X_{n+1} = j \mid X_0 = i_0, ..., X_n = i) = P(f(i, \xi_{n+1}) = j) = P(f(i, \xi_1) = j).$$

■

Example.
Random walk. A particle on $\mathbb{Z}$ starts at 0 and at each step moves right with probability p and left with probability $q = 1 - p$. If $\xi_k = \pm 1$ records the k-th step and $X_n = \sum_{k=1}^{n} \xi_k$, then $X_{n+1} = X_n + \xi_{n+1}$, so $\{X_n\}$ is a homogeneous Markov chain with $p_{i,i+1} = p$ and $p_{i,i-1} = q$.

Example.
Gambler's ruin. The same walk on $\{0, 1, \dots, b\}$, absorbed at the endpoints, is given by $p_{00} = p_{bb} = 1$ and $p_{i,i+1} = p$, $p_{i,i-1} = q$ for $0 < i < b$. Equivalently $X_{n+1} = f(X_n, \xi_{n+1})$ with $f(x, y) = x + y$ for $0 < x < b$ and $f(0, y) = 0$, $f(b, y) = b$.

3.2 Transition Matrix

Definition 3.3. *(State Distribution)*
The distribution of X_n is $\pi_i(n) = P(X_n = i)$, assembled into the row vector $\pi(n) = (\pi_i(n))_{i \in S}$. In particular $\pi(0)$ is the initial distribution.

Proposition 3.2.
The finite-dimensional distributions are determined by $\pi(0)$ and $\mathbb{P}$:

$$P(X_0 = i_0, \dots, X_n = i_n) = \pi_{i_0}(0) p_{i_0 i_1} \cdots p_{i_{n-1} i_n}.$$

Proposition 3.3.
The marginals satisfy $\pi(n) = \pi(n-1)\mathbb{P} = \pi(0)\mathbb{P}^n$.

Proof.
Conditioning on X_{n-1} gives $\pi_j(n) = \sum_i \pi_i(n-1) p_{ij}$. ■

Definition 3.4. *(m-Step Transition)*
The m-step transition probabilities are $p_{ij}^{(m)} = P(X_{n+m} = j \mid X_n = i)$, assembled into the matrix $\mathbb{P}^{(m)}$. In particular $p_{ij}^{(1)} = p_{ij}$ and $p_{ii}^{(0)} = 1$, $p_{ij}^{(0)} = 0$ for $i \neq j$.

Theorem 3.3. *(Chapman-Kolmogorov)*
For all $i, j \in S$ and all $m, n \geq 0$,

$$p_{ij}^{(m+n)} = \sum_k p_{ik}^{(m)} p_{kj}^{(n)}.$$

In matrix form $\mathbb{P}^{(m+n)} = \mathbb{P}^{(m)}\mathbb{P}^{(n)}$, hence $\mathbb{P}^{(m)} = \mathbb{P}^m$.

Proof.
The Markov property at time m yields

$$P(X_{m+n} = j \mid X_0 = i) = \sum_k P(X_m = k \mid X_0 = i) P(X_{m+n} = j \mid X_m = k).$$

■

3.3 Classification of States

Definition 3.5. *(Communication)*
State i leads to j, written $i \to j$, if $p_{ij}^{(n)} > 0$ for some $n \geq 0$. In particular $i \to i$, since $p_{ii}^{(0)} = 1$. The states communicate, denoted by $i \leftrightarrow j$, if $i \to j$ and $j \to i$.

Proposition 3.4.
Communication satisfies the following properties:

- Reflexivity: $i \leftrightarrow i$
- Symmetry: $i \leftrightarrow j$ if and only if $j \leftrightarrow i$
- Transitivity: $i \leftrightarrow j$ and $j \leftrightarrow k$ imply $i \leftrightarrow k$

Therefore communication is an equivalence relation.

Definition 3.6. *(First Passage)*
Starting from $X_0 = i$, the first passage time to j is $\tau_{ij} = \inf\{n \geq 1 : X_n = j\}$, with the convention $\inf \varnothing = +\infty$. The first-passage probabilities are

$$f_{ij}^{(n)} = P\big(\tau_{ij} = n \mid X_0 = i\big) = P(X_n = j, X_1 \neq j, \ldots, X_{n-1} \neq j \mid X_0 = i),$$

and $f_{ij} = \sum_{n=1}^{\infty} f_{ij}^{(n)} = P\big(\tau_{ij} < \infty \mid X_0 = i\big)$. Thus $1 - f_{ij}$ is the probability of never hitting j.

Definition 3.7. *(Recurrence and Transience)*
State i is recurrent if $f_{ii} = 1$ and transient if $f_{ii} < 1$. If i is recurrent, the mean return time is

$$\mu_i = E(\tau_{ii} \mid X_0 = i) = \sum_{n=1}^{\infty} n f_{ii}^{(n)},$$

and i is positive recurrent if $\mu_i < \infty$ and null recurrent if $\mu_i = \infty$.

Definition 3.8. *(Period)*
The period of i is $d(i) = \gcd\Big\{n \geq 1 : p_{ii}^{(n)} > 0\Big\}$. The state is aperiodic if $d(i) = 1$ and periodic if $d(i) > 1$. Return to i is possible only at multiples of $d(i)$. An aperiodic positive recurrent state is ergodic.

Proposition 3.5.
For every $i, j \in S$ and $n \geq 1$,

$$p_{ij}^{(n)} = \sum_{\ell=1}^{n} f_{ij}^{(\ell)} p_{jj}^{(n-\ell)},$$

while $f_{ij}^{(1)} = p_{ij}$ and, for $n \geq 2$,

$$f_{ij}^{(n)} = \sum_{k \neq j} p_{ik} f_{kj}^{(n-1)}.$$

Theorem 3.4. *(Recurrence Criterion)*
State i is recurrent if and only if $\sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty$, and transient if and only if $\sum_{n=1}^{\infty} p_{ii}^{(n)} < \infty$.

Proof.
Write $P(z) = \sum_{n=0}^{\infty} p_{ii}^{(n)} z^n$ and $F(z) = \sum_{n=1}^{\infty} f_{ii}^{(n)} z^n$ for $0 \leq z < 1$, so that $p_{ii}^{(0)} = 1$ and $f_{ii}^{(0)} = 0$. The first-passage convolution gives, for $n \geq 1$,

$$p_{ii}^{(n)} = \sum_{\ell=1}^{n} f_{ii}^{(\ell)} p_{ii}^{(n-\ell)}.$$

Multiplying by z^n and summing therefore yields $P(z) - 1 = P(z)F(z)$, hence

$$P(z) = \frac{1}{1 - F(z)}.$$

As $z \uparrow 1$, monotone convergence gives $F(z) \uparrow f_{ii}$ and $P(z) \uparrow \sum_{n=0}^{\infty} p_{ii}^{(n)}$. Thus $f_{ii} = 1$ if and only if $P(1) = \infty$. $\blacksquare$

Remark.
The same sum is the expected number of visits:

$$\sum_{n=1}^{\infty} p_{ii}^{(n)} = E\left(\sum_{n=1}^{\infty} I_{\{X_n = i\}} \mid X_0 = i \right) = \sum_{n=1}^{\infty} E\big(I_{\{X_n = i\}} | X_0 = i \big).$$

Theorem 3.5. *(Weak Ergodic Theorem)*
The Cesàro matrices

$$\mathbb{L}^{(n)} = \frac{1}{n+1}\sum_{k=0}^{n}\mathbb{P}^k$$

converge to a limit $\mathbb{L}$ satisfying

$$\mathbb{P}\mathbb{L} = \mathbb{L}\mathbb{P} = \mathbb{L} = \mathbb{L}^2.$$

Lemma 3.1. *(Hardy-Littlewood)*
Let $a_n \geq 0$ and let $A(z) = \sum_{n=0}^{\infty} a_n z^n$ converge for $0 \leq z < 1$. Then

$$\lim_{n\to\infty}\frac{1}{n+1}\sum_{k=0}^{n}a_k = \lim_{z\to 1^-}(1-z)A(z),$$

whenever either side exists in $[0, +\infty]$.

Theorem 3.6.
If j is recurrent, then for every $i \in S$,

$$\lim_{n\to\infty}\frac{1}{n+1}\sum_{k=0}^{n}p_{ij}^{(k)} = \frac{f_{ij}}{\mu_j},$$

with the convention $\frac{1}{\mu_j} = 0$ when $\mu_j = \infty$.

Proof.
Let $P_{ij}(z) = \sum_{n=0}^{\infty} p_{ij}^{(n)} z^n$ and $F_{ij}(z) = \sum_{n=1}^{\infty} f_{ij}^{(n)} z^n$. The first-passage convolution yields $P_{ij}(z) = F_{ij}(z)P_{jj}(z)$, and $P_{jj}(z) = \frac{1}{1-F_{jj}(z)}$, so

$$P_{ij}(z) = \frac{F_{ij}(z)}{1 - F_{jj}(z)}.$$

Since j is recurrent, $F_{jj}(1) = 1$ and

$$1 - F_{jj}(z) = \sum_{n=1}^{\infty} f_{jj}^{(n)}(1 - z^n) = (1-z)\sum_{n=1}^{\infty} f_{jj}^{(n)}\sum_{k=0}^{n-1} z^k.$$

Therefore $\frac{1-z}{1-F_{jj}(z)} \to \frac{1}{\mu_j}$ as $z \to 1^-$, while $F_{ij}(z) \to f_{ij}$. The Hardy-Littlewood lemma converts the Abel limit $(1-z)P_{ij}(z) \to \frac{f_{ij}}{\mu_j}$ into the asserted Cesàro limit. ∎

Theorem 3.7.
If i is recurrent of period $d = d(i)$, then $\lim_{n\to\infty} p_{ii}^{(nd)}$ exists and equals $\frac{d}{\mu_i}$.

Proof.
The Cesàro theorem gives $\frac{1}{N+1}\sum_{k=0}^{N} p_{ii}^{(k)} \to \frac{1}{\mu_i}$. The terms $p_{ii}^{(k)}$ vanish unless d divides k. Taking $N = nd$ and using that $\lim_n p_{ii}^{(nd)}$ exists for a recurrent state,

$$\frac{1}{nd+1}\sum_{m=0}^{n} p_{ii}^{(md)} = \frac{n+1}{nd+1}\cdot\frac{1}{n+1}\sum_{m=0}^{n} p_{ii}^{(md)}.$$

The left side tends to $\frac{1}{\mu_i}$ and $\frac{n+1}{nd+1} \to \frac{1}{d}$, so the Cesàro means of $p_{ii}^{(md)}$ tend to $\frac{d}{\mu_i}$. A convergent sequence has the same limit as its Cesàro means. ■

Corollary 3.1.

- i is transient if and only if $\sum_n p_{ii}^{(n)} < \infty$;
- i is null recurrent if and only if $\sum_n p_{ii}^{(n)} = \infty$ and $p_{ii}^{(n)} \to 0$;
- i is positive recurrent if and only if $\limsup_n p_{ii}^{(n)} > 0$.